

Problem 8.17. Show that if every NP-hard language is also PSPACE-hard, then $\text{PSPACE} = \text{NP}$.

Proof. Assume that every NP-hard language is also PSPACE-hard. This implies that SAT is also PSPACE-hard. In other words, every language in PSPACE is polynomial time reducible to SAT . Then, every language A in PSPACE can be decided in non-deterministic polynomial time by first converting an instance of A to an instance of SAT , which can be decided in non-deterministic polynomial time. Therefore, $\text{PSPACE} = \text{NP}$. $\square$